

Basic Principles of Medicine 1
Module: Foundation to Medicine | Lecture 2

Histology of Cellular Organelles

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Objectives

Discuss the nucleus as a major organelle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0738
Give the function(s) of the nucleolus.	SOM.MK.I.BPM1.1.FTM.3.HCB.0739
Describe the microscopic structure of the nucleolus.	SOM.MK.I.BPM1.1.FTM.3.HCB.0740
State the three components of chromatin.	SOM.MK.I.BPM1.1.FTM.3.HCB.0741
Describe how these three components come together to form chromatin.	SOM.MK.I.BPM1.1.FTM.3.HCB.0742
Distinguish between euchromatin and heterochromatin in a nucleus.	SOM.MK.I.BPM1.1.FTM.3.HCB.0743
Predict cellular activity based on the nuclear chromatin structure.	SOM.MK.I.BPM1.1.FTM.3.HCB.0744
Give the function of the nuclear envelope.	SOM.MK.I.BPM1.1.FTM.3.HCB.0745
Describe the structure of nuclear envelope including the nuclear pore complex.	SOM.MK.I.BPM1.1.FTM.3.HCB.0746
Differentiate between the following organelles based on their microanatomical structures and function: ribosomes, endoplasmic reticulum (RER & SER), Golgi apparatus, lysosome, peroxisome and mitochondria.	SOM.MK.I.BPM1.1.FTM.3.HCB.0747
Describe the bi-directional vesicle transport between the endoplasmic reticulum and Golgi apparatus.	SOM.MK.I.BPM1.1.FTM.3.HCB.0748
Describe protein trafficking for lysosomes, secretion, plasma membrane, nucleus, mitochondria, cytosol and peroxisomes.	SOM.MK.I.BPM1.1.FTM.3.HCB.0749
Describe the three cellular pathways to lysosomal degradation.	SOM.MK.I.BPM1.1.FTM.3.HCB.0750
Identify the consequences of dysfunctional peroxisomes for example Zellweger syndrome.	SOM.MK.I.BPM1.1.FTM.3.HCB.0751
Identify the causes and consequences of lysosomal storage diseases (for example Tay-Sachs) based on the defective enzyme, accumulated products and organs most susceptible.	SOM.MK.I.BPM1.1.FTM.3.HCB.0752

Objectives

Define autophagy	SOM.MK.I.BPM1.1.FTM.3.HCB.0753
Describe the process of autophagy including the coordinated process from the membrane conformational changes to fusion with lysosomes (autolysosomes)	SOM.MK.I.BPM1.1.FTM.3.HCB.0754
Discuss the significance of autophagy	SOM.MK.I.BPM1.1.FTM.3.HCB.0755
Give the function(s) of the proteasome	SOM.MK.I.BPM1.1.FTM.3.HCB.0756
Describe the structure of the proteasome	SOM.MK.I.BPM1.1.FTM.3.HCB.0757
Describe the four types of basic tissue types	SOM.MK.I.BPM1.1.FTM.3.HCB.0759
Identify the origin of the four basic types of tissue	SOM.MK.I.BPM1.1.FTM.3.HCB.0760

Recommended Reading

- *Material covered in lecture handout

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 2: Cell Cytoplasm Pg. 62-81,

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=257424028&bookid=3290>

Chapter 3: The Cell Nucleus Pg. 82-105

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=257424454&bookid=3290>

Lippincott's Illustrated Reviews: Cell & Molecular Biology 3e

Chandler & Viselli 2024

Chapter 5

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=255703802&bookid=3244>

Important note

Some of the slides provided in this handout will have a green background

This is information that will help you understand the lecture material but will NOT be covered during the lecture delivery

Terminology

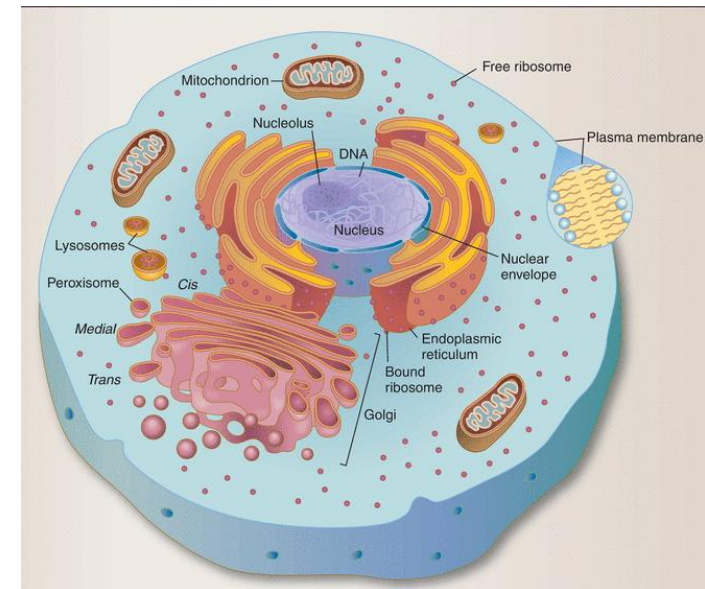
- **Cytoskeleton** – a network of proteins that provide scaffolding or allow movement for the structures inside the cell
- **Polymerization** – the process by which cytoskeletal filaments are formed and lengthened through assembly of their protein subunits
- **Treadmilling** – continual process of assembly on one end and disassembly on the other to maintain the optimal length of the cytoskeletal filament
- **Mitosis** – cell division resulting in two cells identical to the original
- **Axoneme** – the central portion (core) of cilia, flagella and primary cilia
- **Apical surface of cell** – the portion of the cell that is not attached to anything also referred to as the free surface
- **Basal surface of cell** – the portion of the cell that is attached to the basement membrane
- **Lateral surfaces of cell** – the surfaces of the cell that are adjacent to other cells
- **Terminal web** – a network of actin filaments that sit just deep to the plasma membrane
- **Signaling molecule** – a substance that can be recognized by cell surface receptors and serve as a way for cells to communicate with each other
- **Receptor** – a protein that is capable of binding to a signaling molecule. May be part of the cell membrane or in the cytosol

Eukaryotic cell – overview

- **Plasma membrane** - Outermost organelle which keeps everything together, also surrounds most organelles
- **Cytoplasm** – Everything on the intracellular aspect of the plasma membrane
 - Cytoplasmic matrix (**cytosol**) – aqueous gel containing molecules of different size where the physiological processes of the cell takes place
 - **Cytoskeleton** – polymerized proteins that serve as a framework for cell structure and organization, allows for movement within the cell and the cell itself. Determines the fluidity of the cytosol and plays an important part in cell division
- **Organelles** – little organs
- **Inclusions** – accumulations in cells byproducts of metabolism

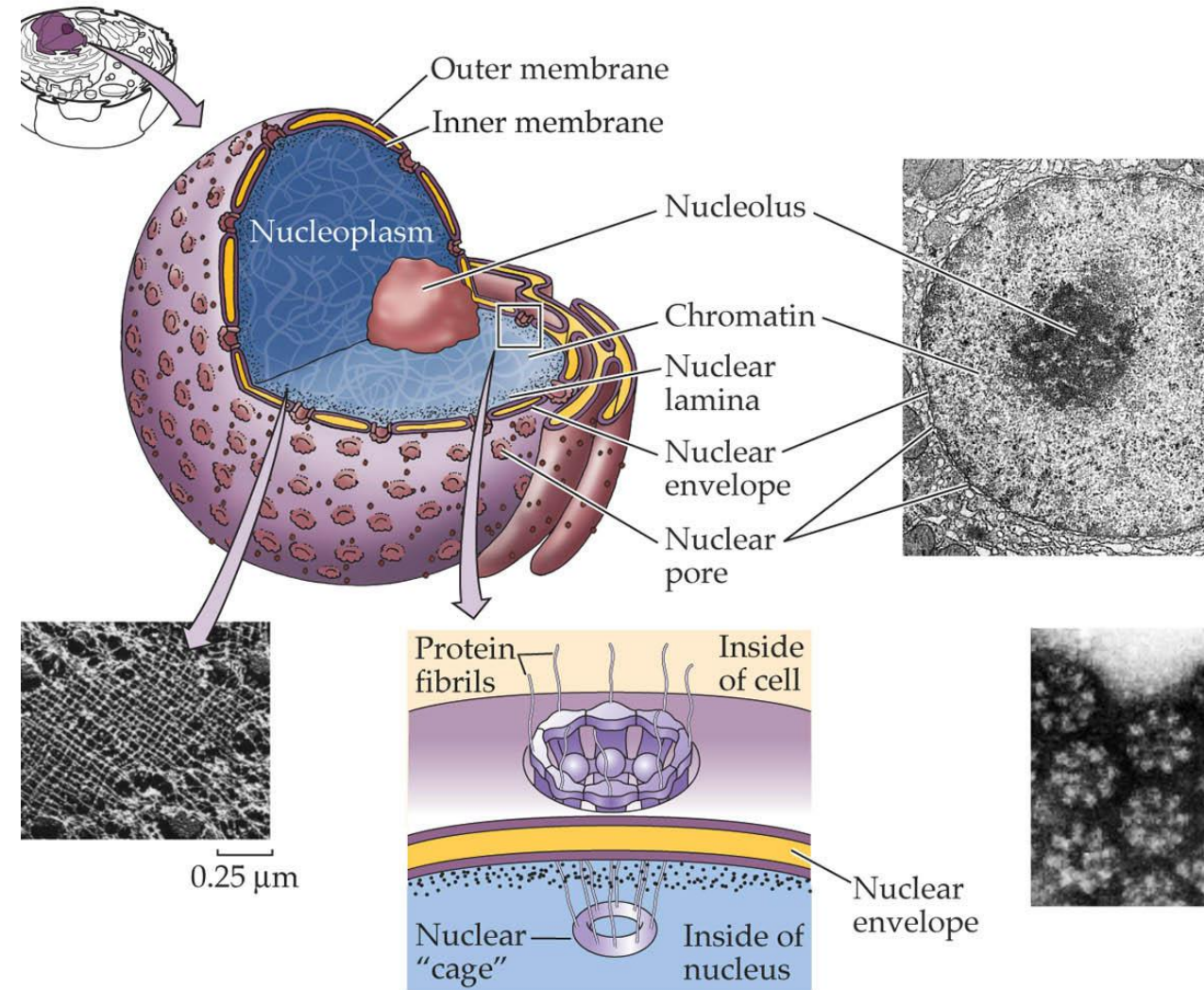
Cytoplasmic Organelles

- Functionally distinct intracellular locations where processes necessary for life occur
- Most are membrane enclosed
 - The membrane is very similar to the plasma membrane with exclusion of cholesterol
- The organelles are arranged in the cytosol in a very particular way to facilitate their function
 - They are interconnected and joined by the cytoskeleton - **do not float freely**
- Each organelle has a specific function, but they often work together to complete a complex task
 - Organelles necessary for gene expression and protein synthesis in various forms
 - Nucleus
 - Ribosomes
 - Endoplasmic reticulum
 - Golgi complex
 - Energy production organelles
 - Mitochondria
 - Digestion and detoxification
 - Lysosomes and peroxisomes



Cell Nucleus

- Largest organelle in the cell
 - ~10% cell volume
 - Contains **genetic information** and machinery for DNA replication and RNA transcription
- In non-dividing (interphase) cells the nucleus contains:
 - Nuclear envelope (membrane)
 - Nuclear pores
 - Nuclear lamina
 - Nucleoplasm
 - Chromatin – DNA and proteins
 - Nucleolus

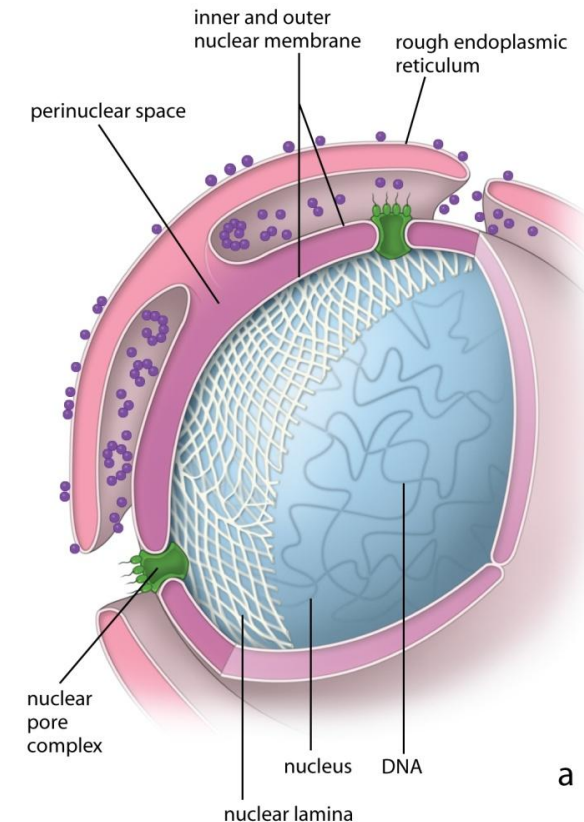
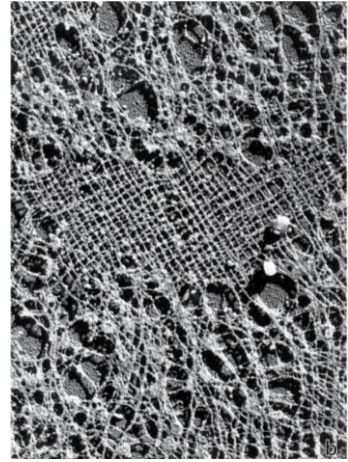


LIFE: THE SCIENCE OF BIOLOGY, Seventh Edition, Figure 4.9 The Nucleus Is Enclosed by a Nuclear Envelope
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Cell Nucleus – nuclear lamina and envelope

Nuclear Envelope (membrane)

- Inner Nuclear Membrane
 - Faces the nucleoplasm
 - In contact with the nuclear lamina
- Outer Nuclear Membrane
 - Faces the cytoplasm
 - Continuous with endoplasmic reticulum (ER)
 - Cytoplasmic surface contains ribosomes

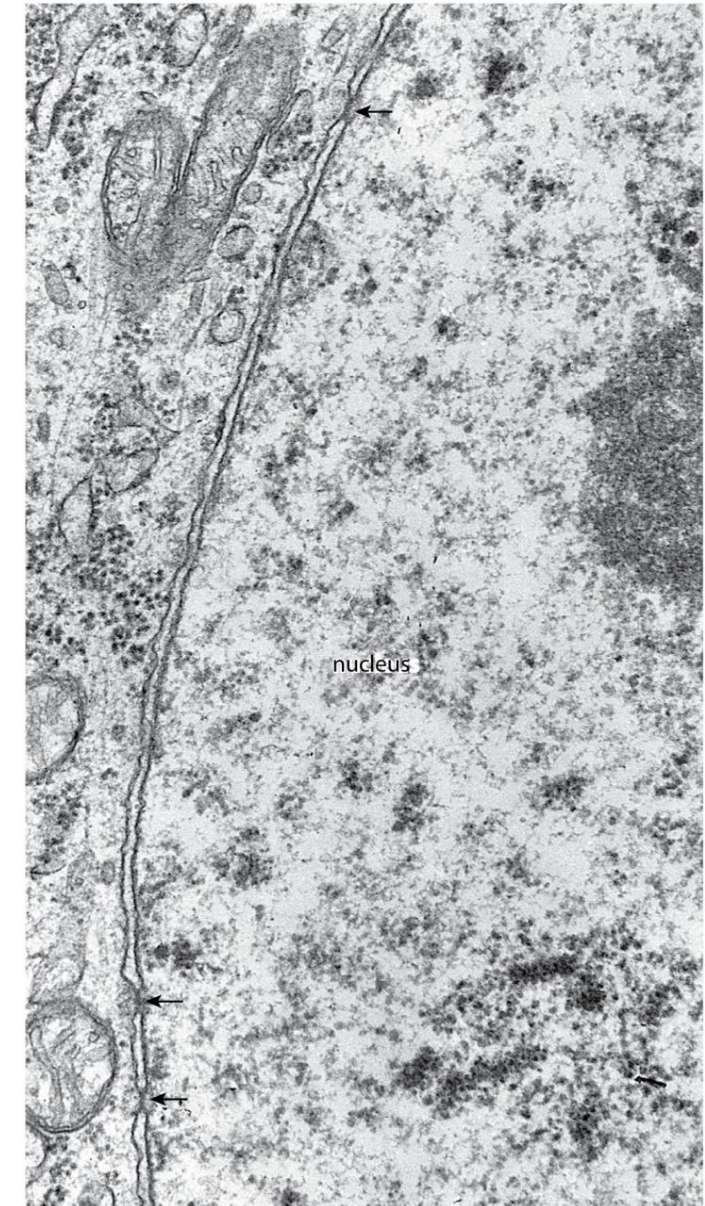
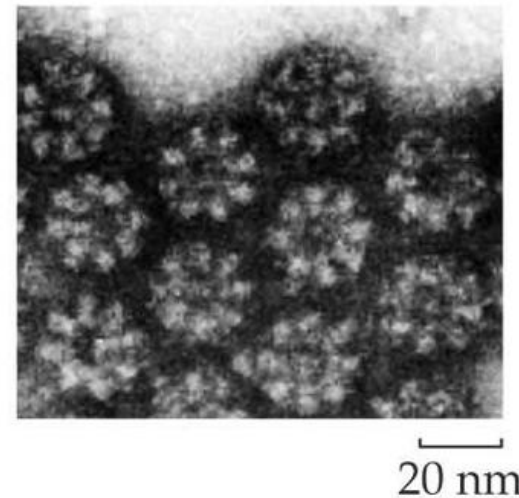
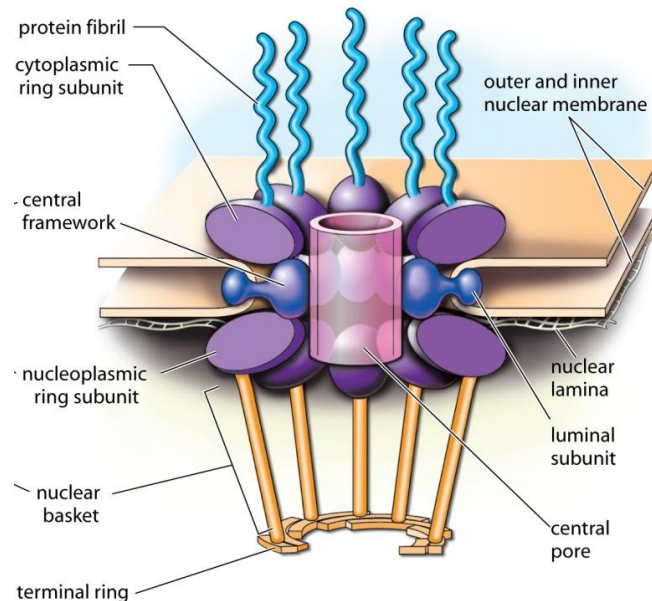


Nuclear Lamina

- Thin, sheet-like meshwork (**Lamins A & B**)
- Scaffold for nuclear envelope, chromatin & nuclear pores
- Disassembles during mitosis

Cell Nucleus – Nuclear Pore Complex (NPC)

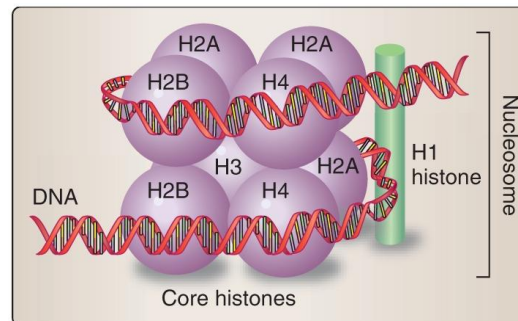
- Allows for selective **transport** between the nucleus and cytoplasm
 - Ions and molecules **< 9nm** diffuse freely through
 - Molecules **> 9nm** must be transported by active transport
- Span the inner and outer nuclear membranes
 - 80-100nm in diameter
- Made up of more than 50 proteins
 - Nucleoporins



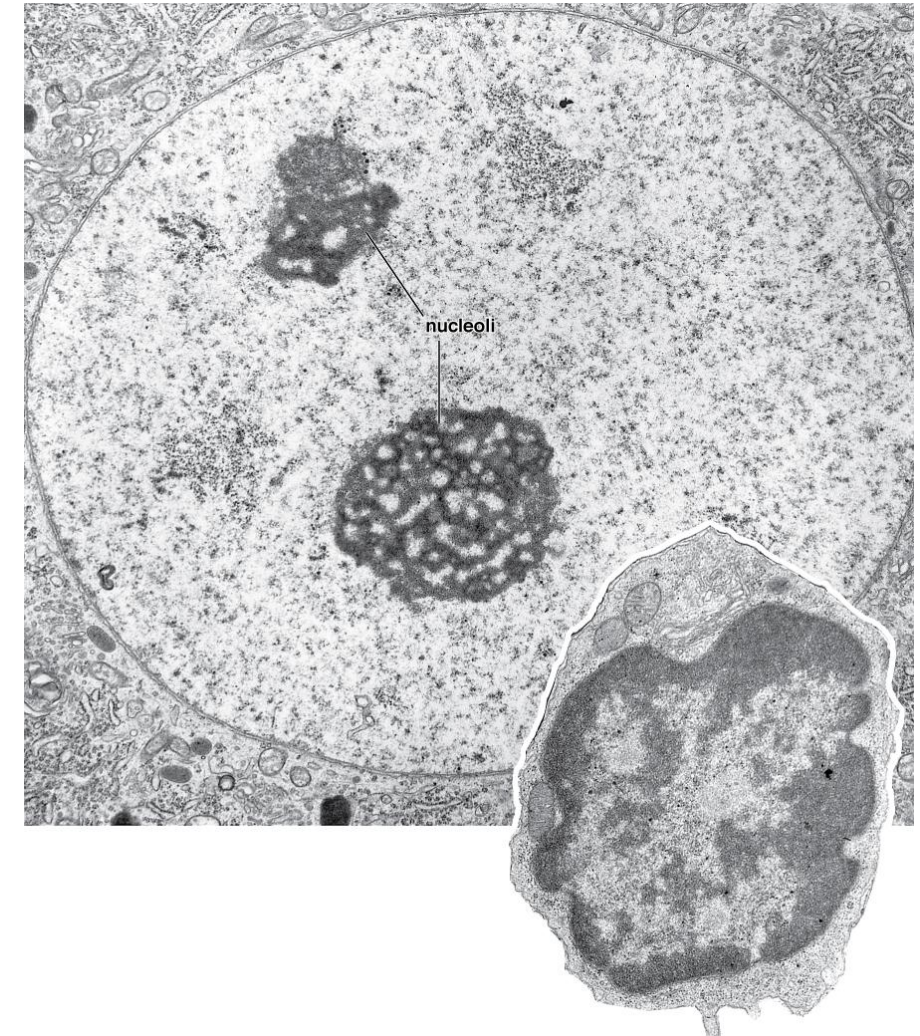
Cell Nucleus – Chromatin

DNA - 2.85 billion base pairs if when stretched out measures 1.8m, which is 100,000 times the diameter of the average nucleus

- **Chromatin:** DNA folded in an organized manner
- Dispersed areas (lightly stained) of **euchromatin**
- Highly condensed (darkly stained) areas of **heterochromatin**
- Most cells have a heterogenous chromatin pattern
- Smallest folded unit is called a **nucleosome**
 - 2 loops of DNA wrapped around core of **8 histone** proteins
 - “beads on a string” formation

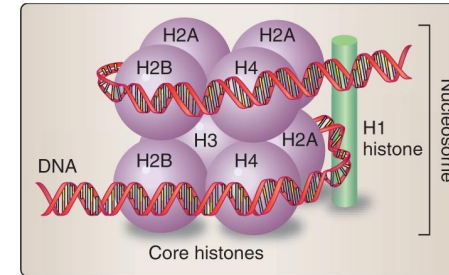


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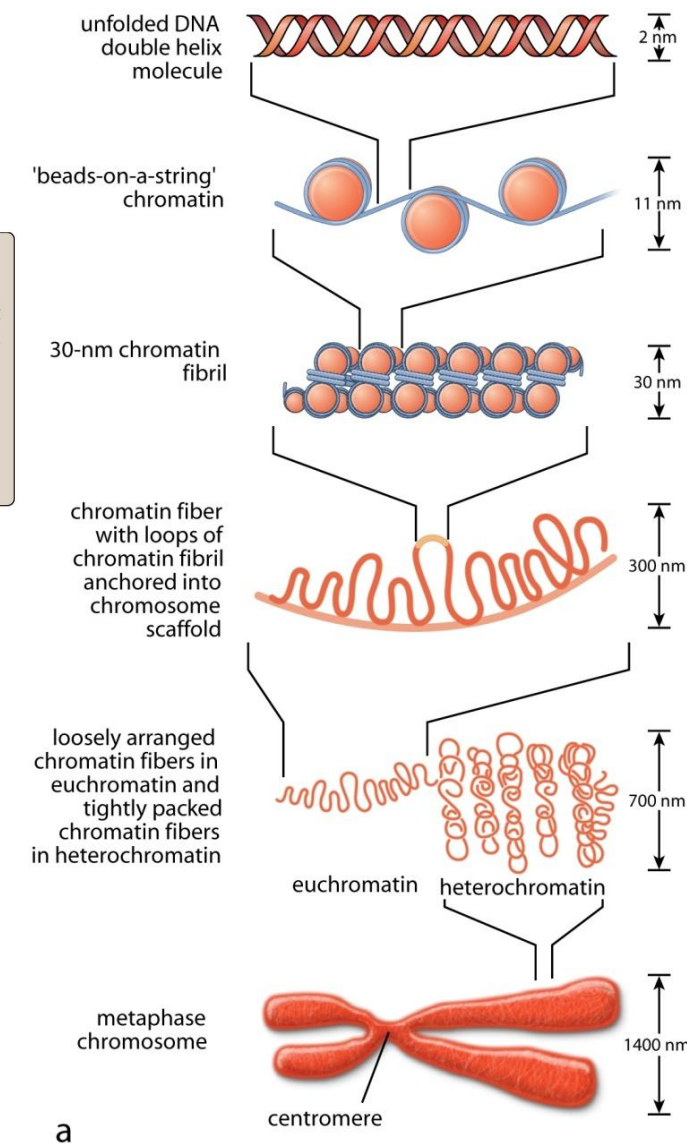


Cell Nucleus – Chromatin folding

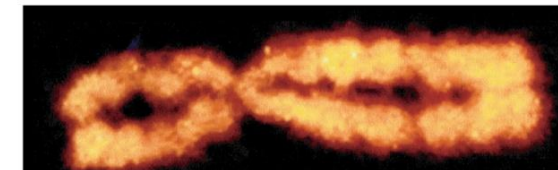
- 6 nucleosomes form a chromatin fibril = **coil**
 - Coils are looped and held together by **nonhistone proteins**
- During cell division, condenses into **chromosomes**
 - Chromatids joined at the **centromere**
 - Each cell (excluding the gametes) contain **23 pairs**
 - 22 autosomes and 2 sex chromosomes
 - The ends of the chromosome are called telomeres, which shorten with each cell division
 - **Telomere** length is an indication of the lifespan of the cell



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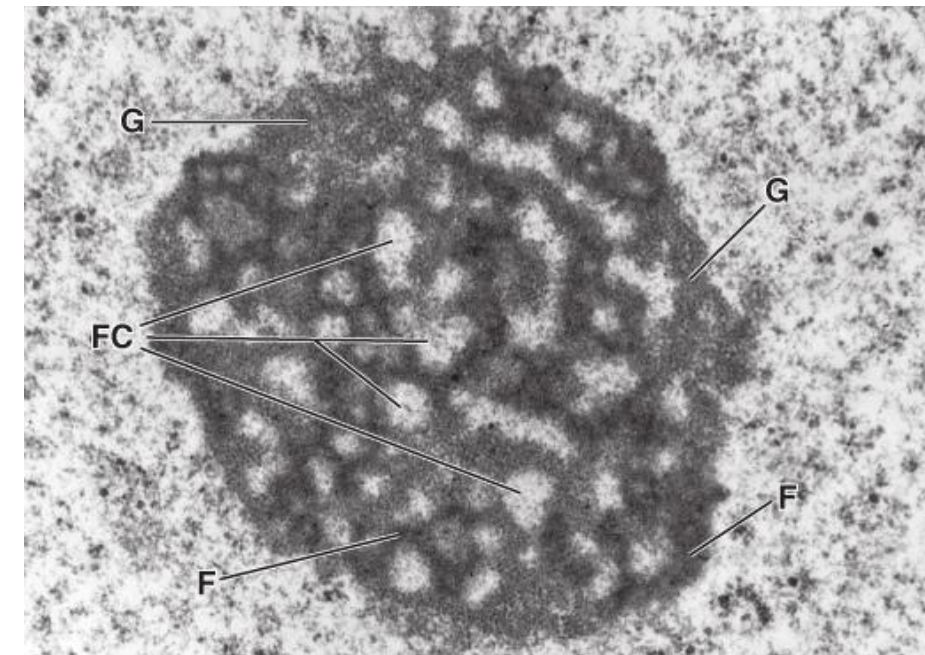


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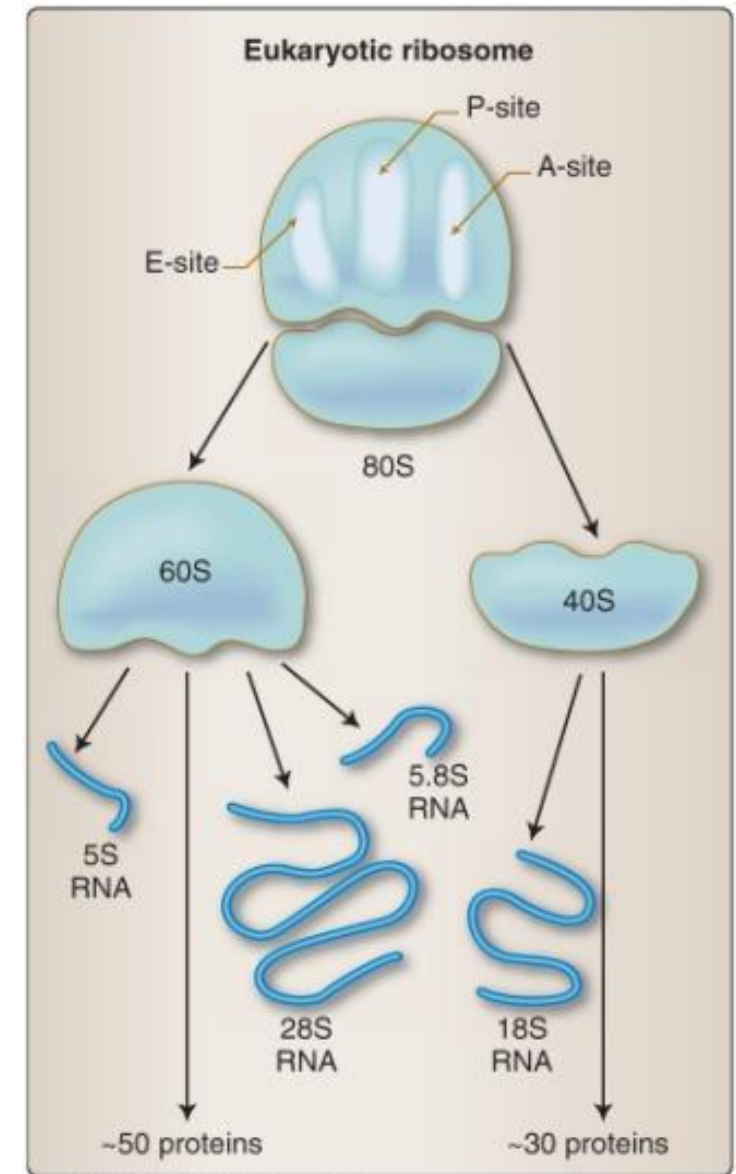
Cell Nucleus – Nucleolus

- Non-membranous region of the nucleus
- Primary site for **ribosomal production** and **assembly**
 - Contains transcriptionally active rRNA genes
- Varies in size but well developed in cells involved actively in protein synthesis
 - May be duplicated
- Three morphologically distinct regions
 - **Fibrillar center (FC)** - pale staining
 - DNA loops of specific chromosomes
 - rRNA genes, RNA polymerase I and transcription factors
 - **Fibrillar material (F)** – dark staining
 - Ribosomal genes in active transcription
 - rRNA
 - **Granular material (G)** – in between staining
 - Initial ribosomal assembly



Ribosomes

- Make proteins
- Consist of 2 ribosomal subunits and associated proteins
 - Assemble to synthesize a protein
 - Disassemble after
 - Binds **mRNA**
 - Translates its
 - Assembles the protein
- Polyribosome
 - Many ribosomes bound to a single mRNA molecule
- May be **membrane-bound or free**
- Two pathways for protein synthesis
- **Ribosomopathies**
 - e.g. Treacher Collins Syndrome



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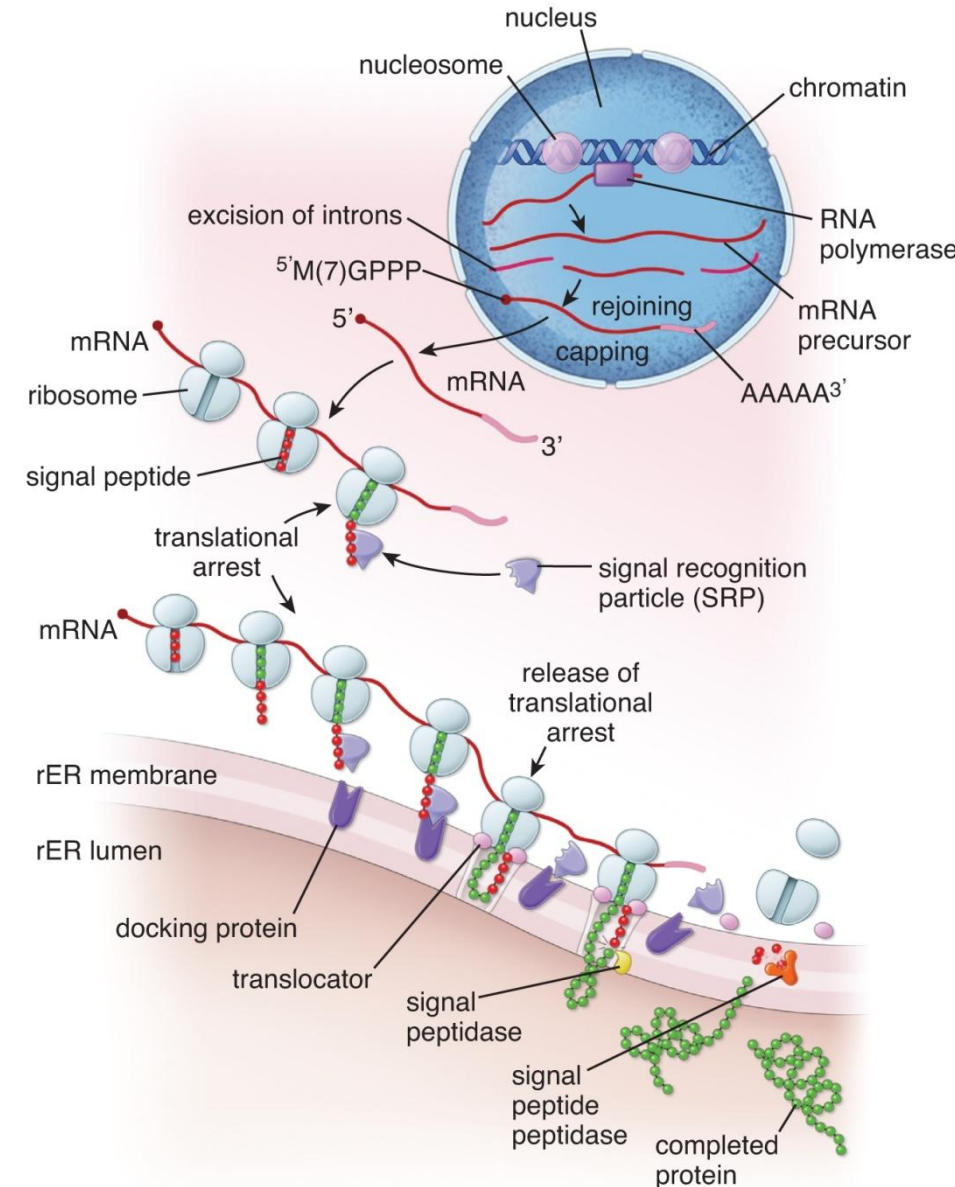
Ribosomes - Protein assembly and Trafficking

Membrane-bound ribosomes

- Attached to cytoplasmic surface of the ER membrane
 - Rough ER (RER)
- Requires an ER **signal sequence on mRNA**
- Synthesize **proteins that are being translocated into the ER lumen**

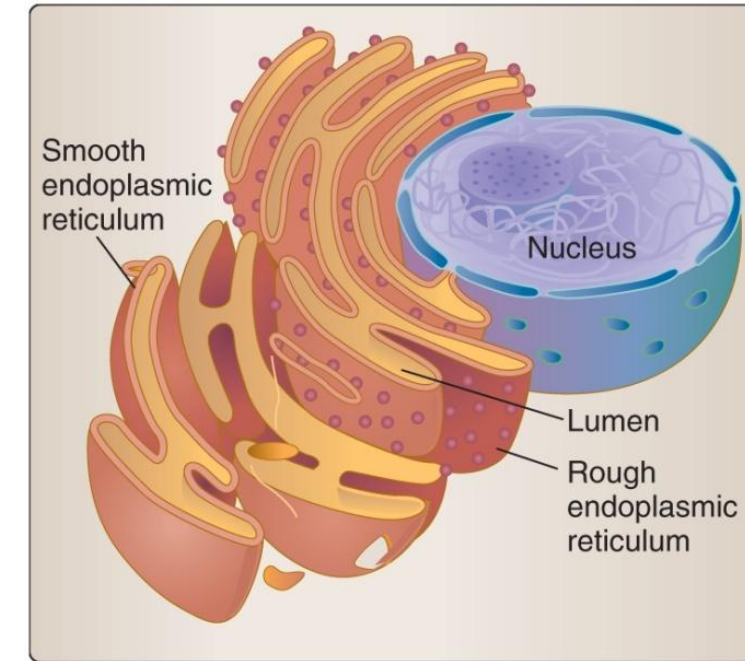
Free Ribosomes

- Not attached to any membrane
 - mRNA does not have ER signal sequence
- Synthesize **all other proteins**



Endoplasmic Reticulum (ER)

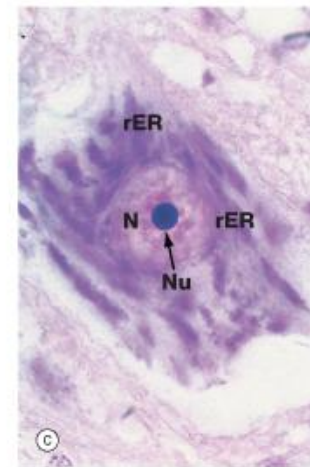
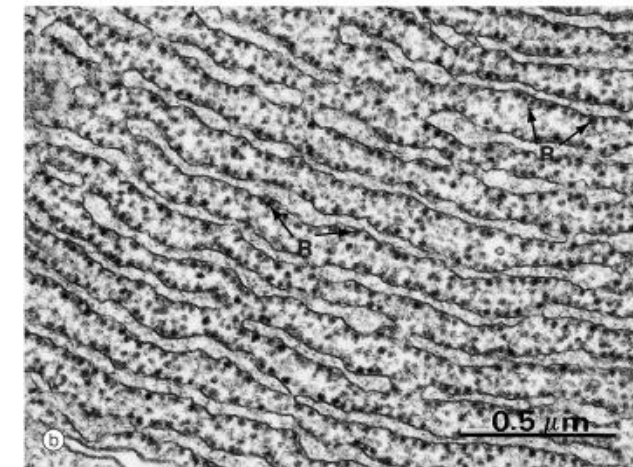
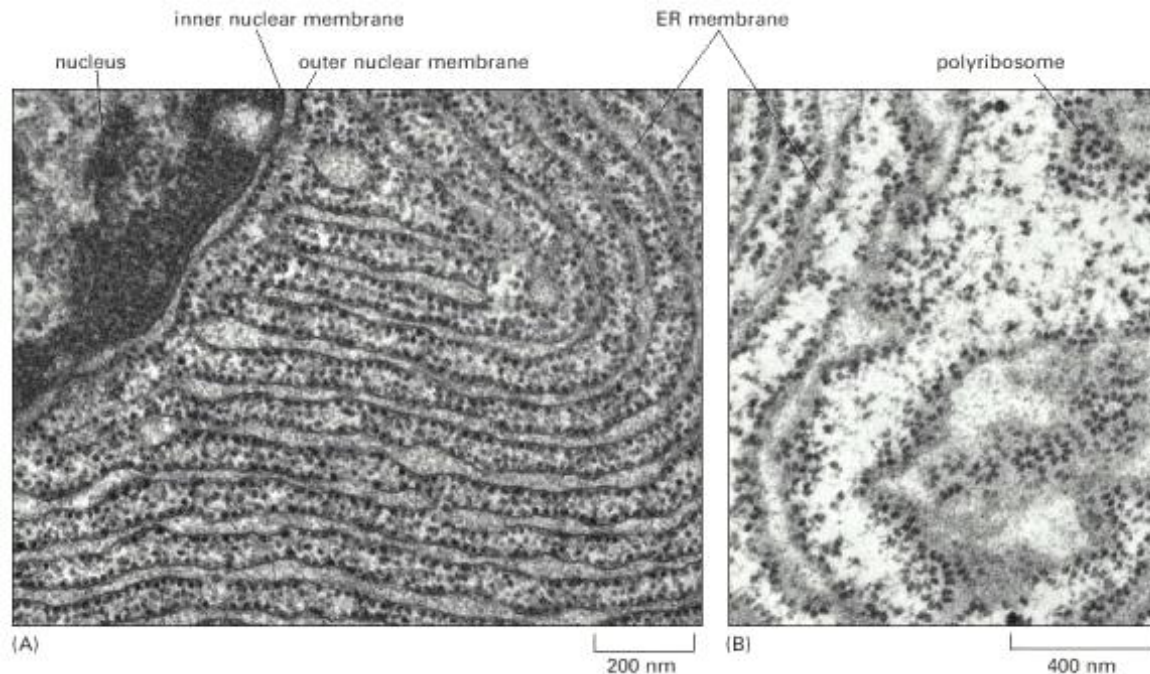
- Interconnected network of branching tubules & flattened sacs distributed throughout the cytoplasm
- At certain sites, the ER membrane is continuous with the outer nuclear envelope membrane
- Two regions
 1. **Smooth ER** functions in synthesis of lipids & detoxification
 2. **Rough ER** functions in synthesis of proteins destined for plasma membrane, lysosomes or secretion
- **ER lumen**
 - Single, continuous, internal space
 - Separated from cytoplasm by the ER membrane



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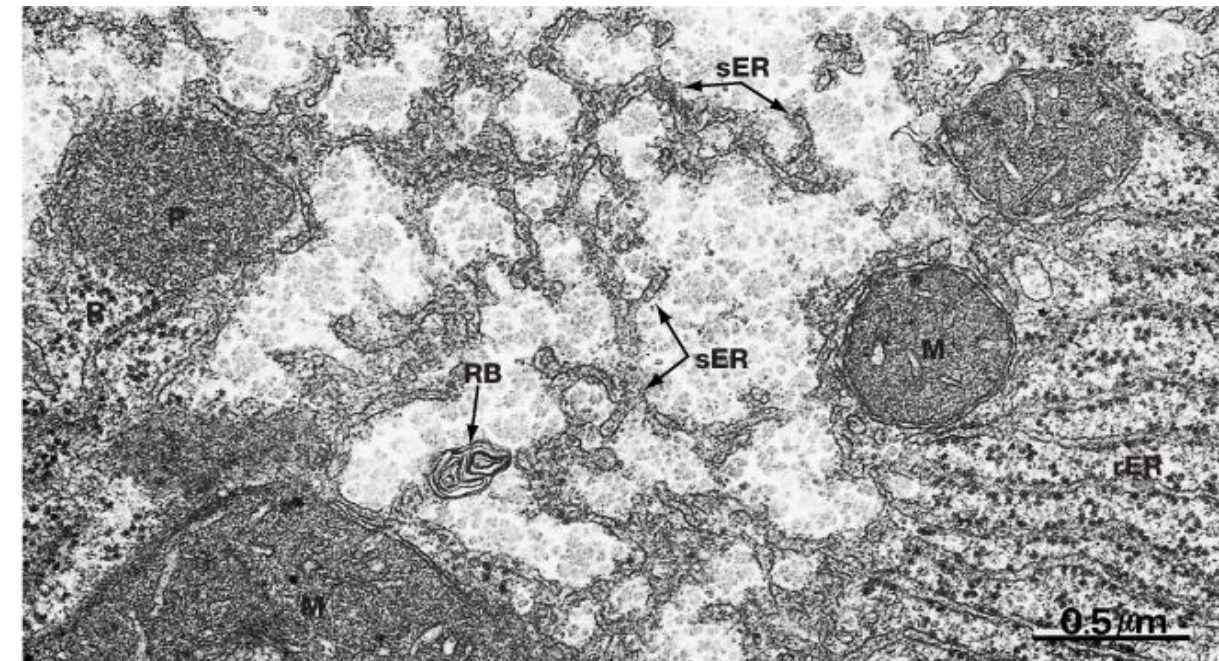
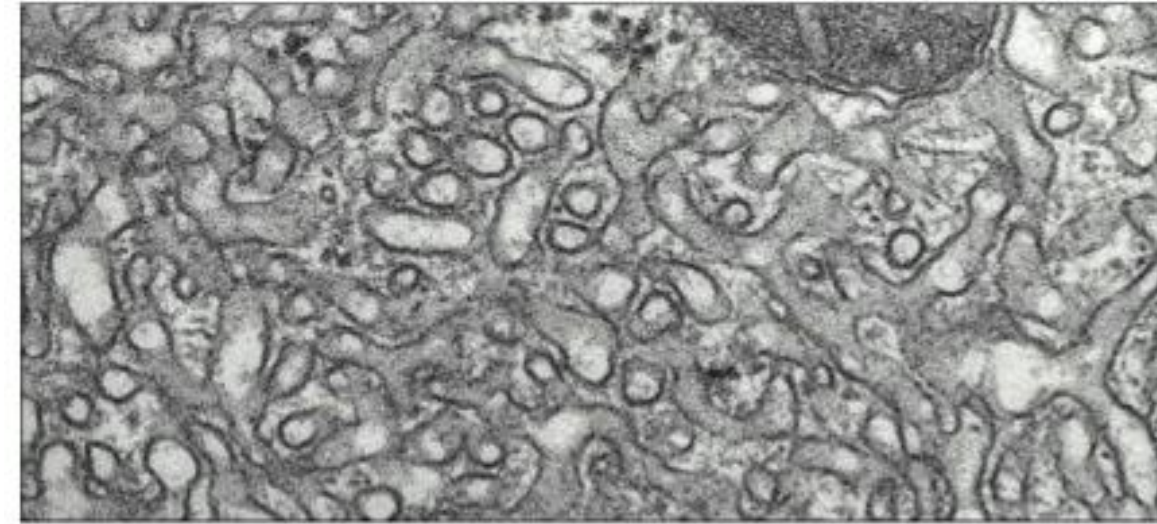
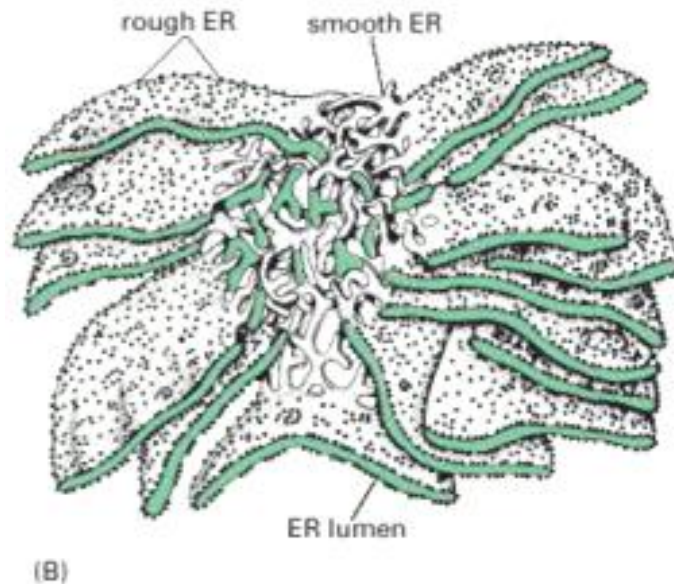
Rough Endoplasmic Reticulum (rER)

- Region with ribosomes attached to cytoplasmic surface of the ER
- Only mRNA molecules with an ER signal sequence bind to rER membranes
- Abundant in cells specialized in **protein synthesis**
 - Secretory cells producing proteins for extracellular export
- Site of initial post-translational modifications & folding



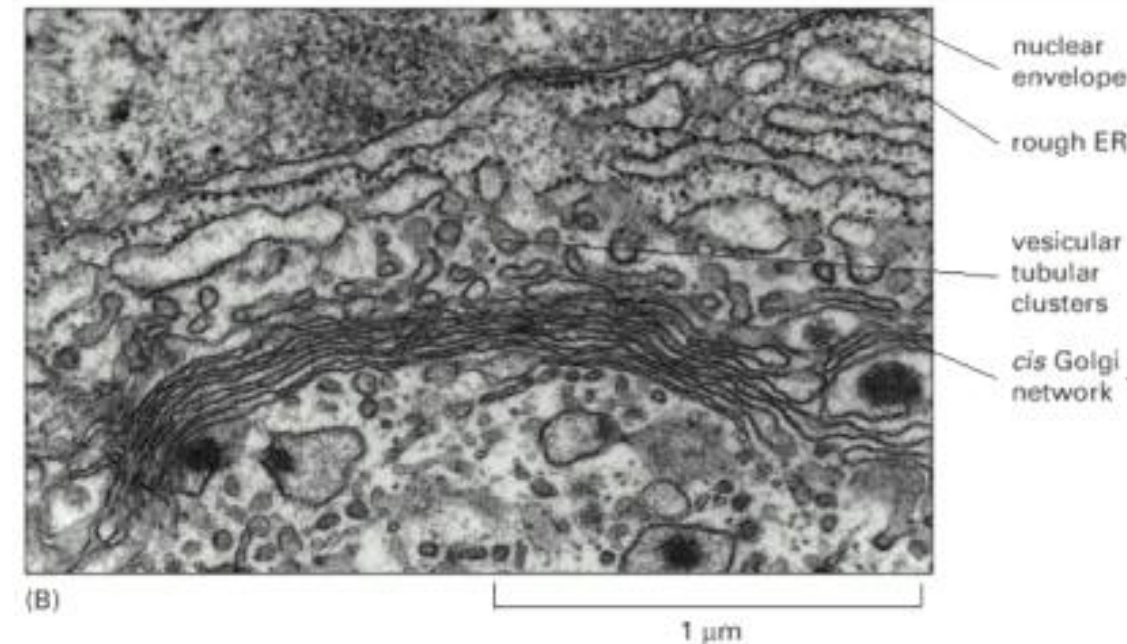
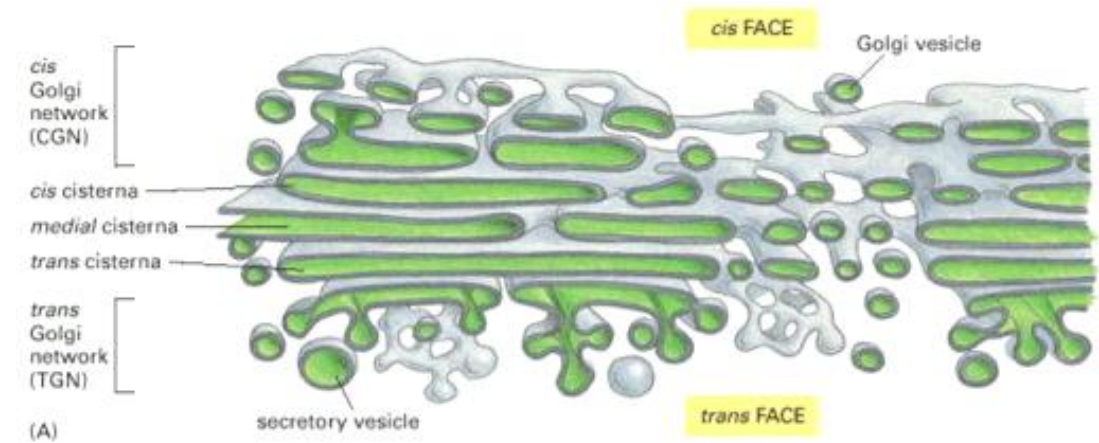
Smooth Endoplasmic Reticulum (sER)

- Region of the ER that lacks bound ribosomes
- Abundant in cells specialized in **lipid metabolism**
 - Well-developed in cells that synthesize and secrete steroids
- Plays a major role in detoxification
 - i.e. **cytochrome P450 system**
- Also functions to sequester calcium
 - Muscle cells (sarcoplasmic reticulum)



Golgi Apparatus

- Makes **vesicles**
- Especially well-developed in **secretory cells**
- Usually located next to the nucleus and centrosome
- Complex of flattened, **membrane-enclosed cisternae**
- Each Golgi stack has 2 faces
 - Close to the rER
 - **Cis Golgi Network (CGN)**
 - Away from rER
 - **Trans Golgi Network (TGN)**
- Exit of mature vesicles



Golgi Apparatus - Functions

- Vesicles travel from cisterna to cisterna as they bud off and fuse with the next stack in the required direction
 - Post-translational modification
 - Each region is responsible for a specific modification
 - Changes depend on destiny
- Sorting
- Packaging
- Produce the precursor vesicles for lysosomes

Post-translational modification

Glycosylation—Addition of carbohydrate

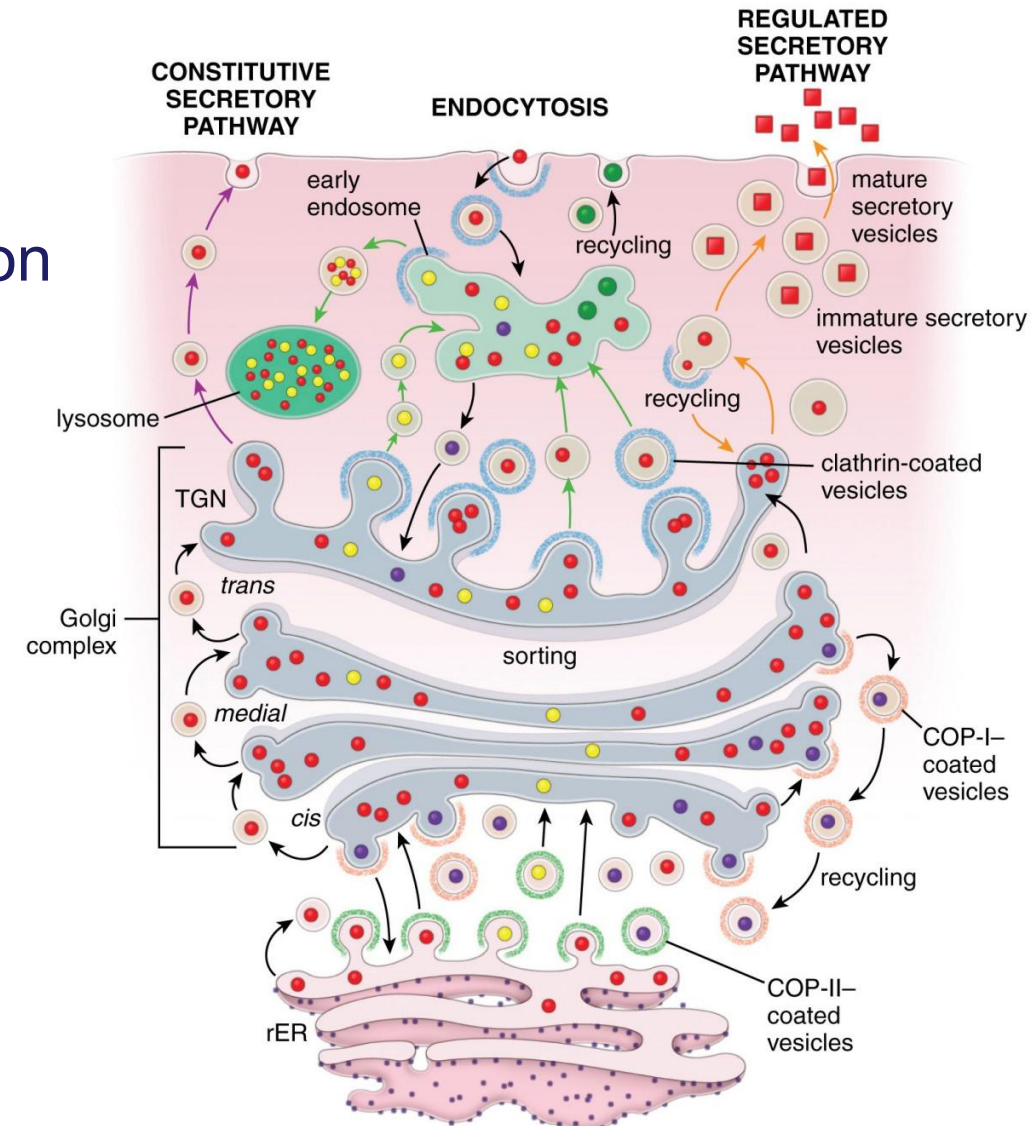
Sulfation—Addition of sulfur

Phosphorylation—Addition of phosphate

Proteolysis—Cleavage of peptide bonds

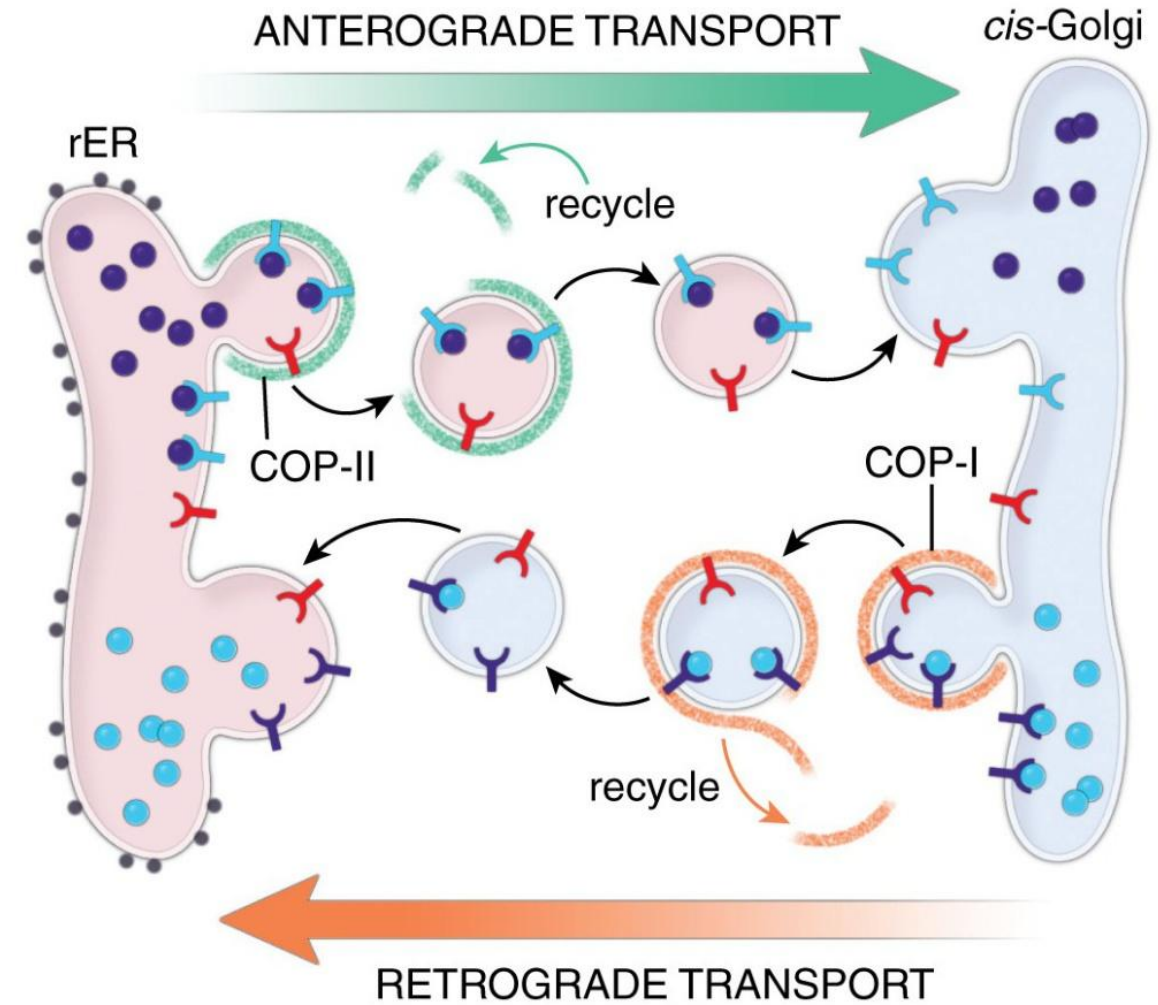
Golgi Apparatus – Vesicular trafficking

- **Constitutive Secretory Pathway**
 - Basolateral or Apical
 - Vesicles produced for continuous secretion
- **Regulated Secretory Pathway**
 - Vesicles are stored until needed
 - **A stimulus is required for release**
- **Lysosomal pathway**
 - Phosphorylation on attached manose
 - **Manose-6-phosphate**
 - Released as a **clathrin coated transport vesicle**



rER & Golgi Transport

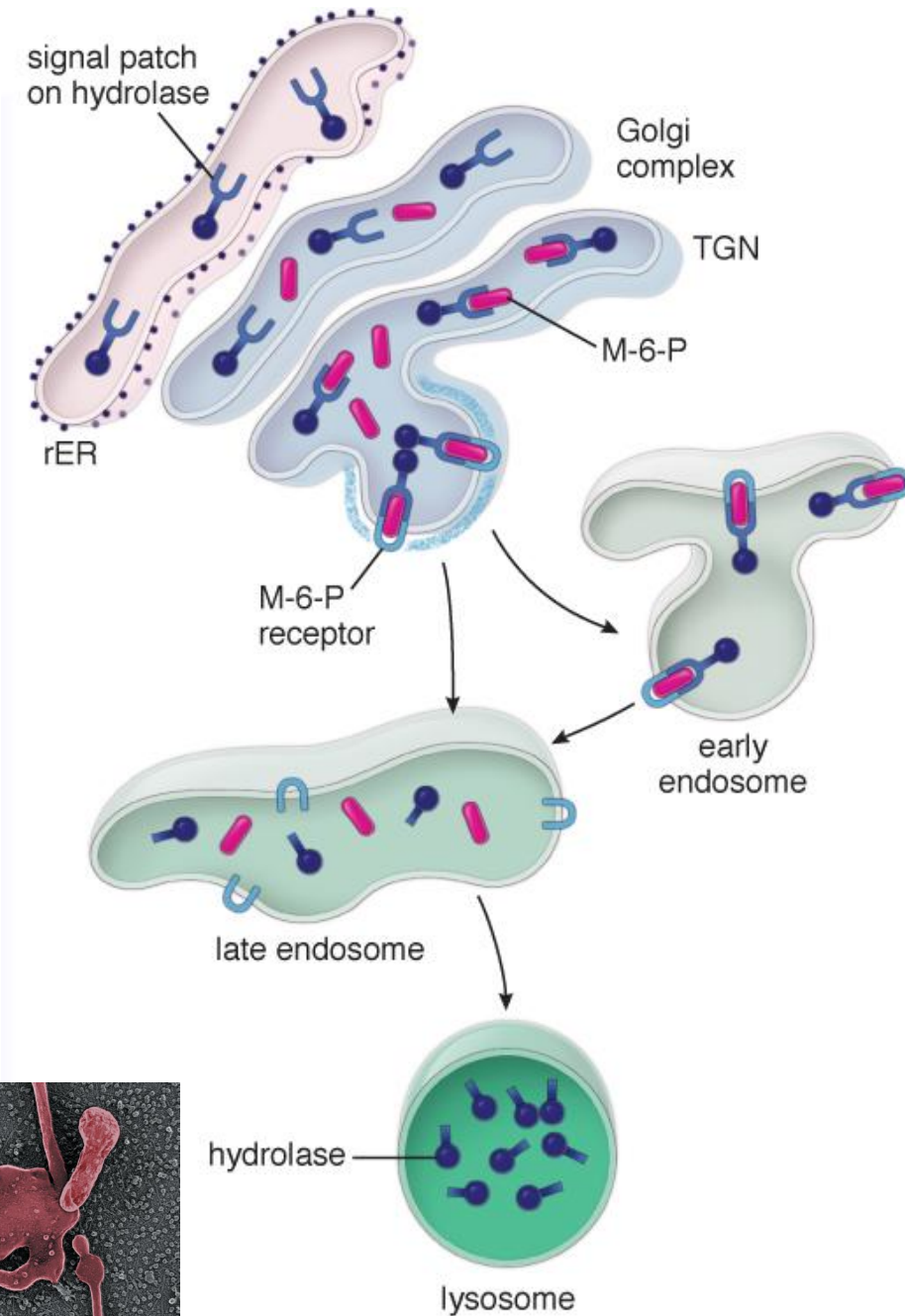
- Bidirectional
 - Anterograde – rER to cis-Golgi network
 - Retrograde – cis-Golgi network to rER
- Mediated by **coatomers** (COPs)
 - COP-I mediates retrograde transport
 - COP-II mediates anterograde transport



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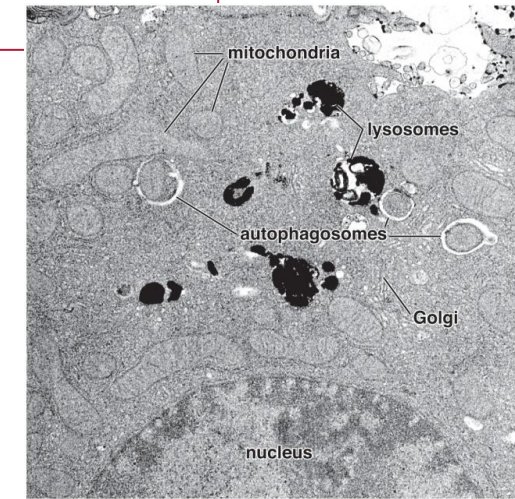
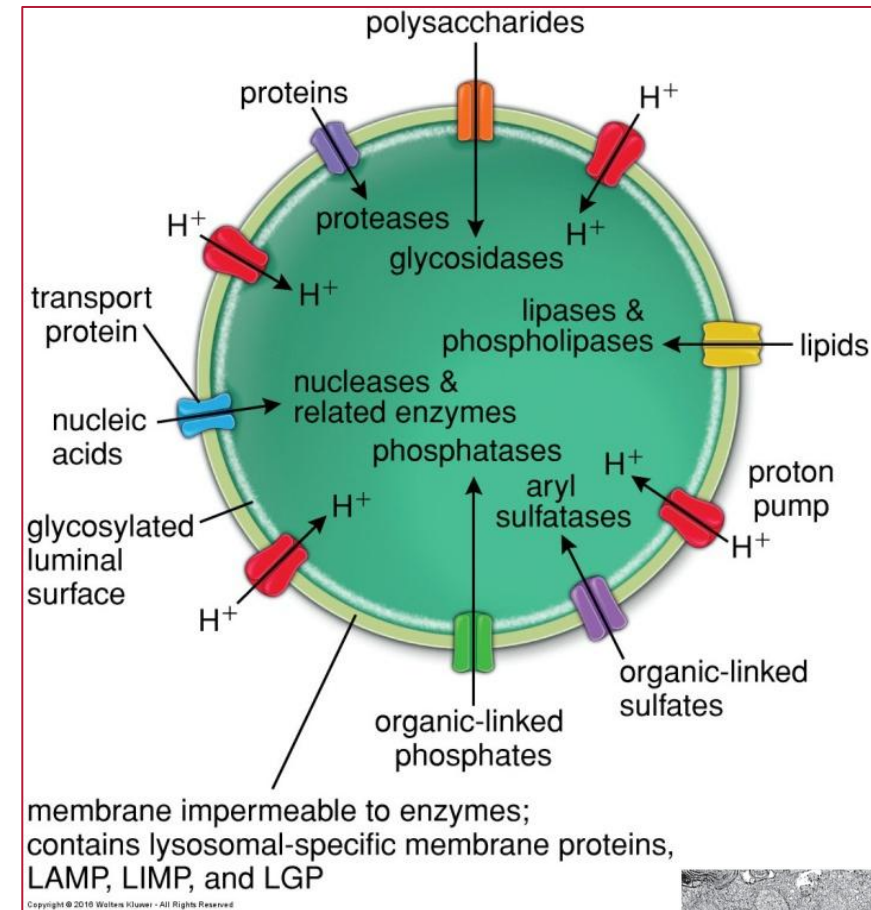
Endosomes

- Membrane-bounded compartments
- Form part of endocytotic pathways
- May contain membrane proteins, fluids, signaling molecules etc.
- Functions to sort and recycle proteins
- Redirects them to different cellular compartments
- Communicates with **RER** for continuous production of hydrolases
- Fuses with a **Manose-6-phosphate** containing vesicle = **Lysosome**



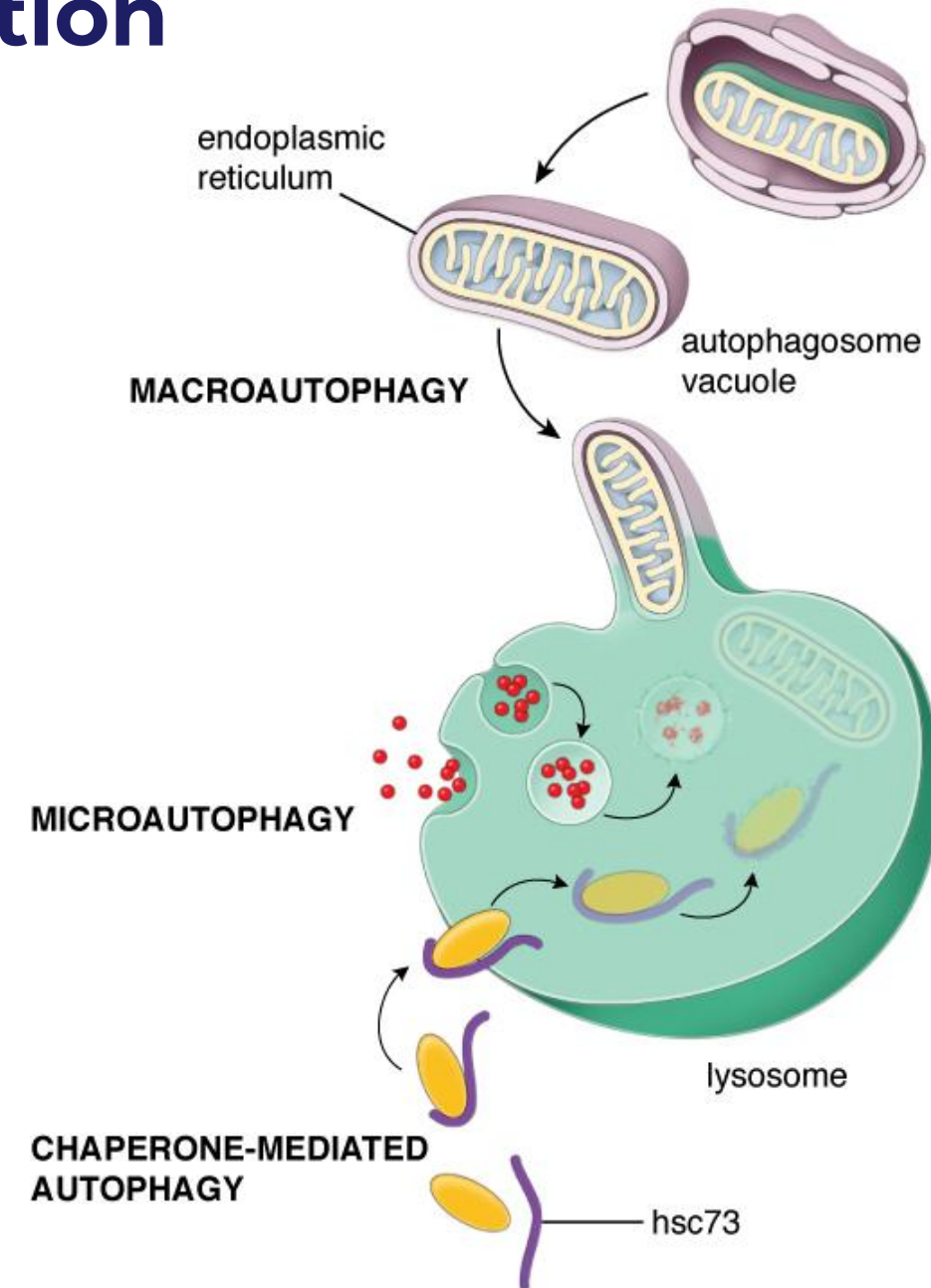
Lysosomes

- Formed from fusion of endosome or phagosome with **Manose-6-phosphate (M6P)** containing transport vesicle
- Digestive organelles
 - **controlled intracellular digestion**
- Filled with ~ **40 types of hydrolytic enzymes**
 - Acid hydrolases
 - Require an acidic environment
 - low pH (~4.7)
 - Specialized based on cell type



Lysosomes – pathways to digestion

- Extracellular large particles (**Phagocytosis**)
 - Bacteria, cell debris
- Extracellular small particles (**Pinocytosis**)
 - Receptor mediated endocytosis
- Intracellular particles (**Autophagy**)
 - Cellular pathway to degrade cellular proteins & organelles
 - Maintains **homeostasis** of the cell by balancing the anabolic and catabolic activity
 - Intracellular membrane (ER) surrounds the organelle and cytoplasm: **Autophagosome**
 - Fusion of autophagosome with lysosome
 - Contents are degraded, recycled and reused



Lysosomal Storage Diseases (LSDs)

- Dysfunctional lysosomes (>40 LSDs)
- Often result from mutations in genes that encode lysosomal enzymes
- Results
 - Accumulation of undigested products
 - Disruption of normal cell function
 - Cell death

Tay-Sachs Disease

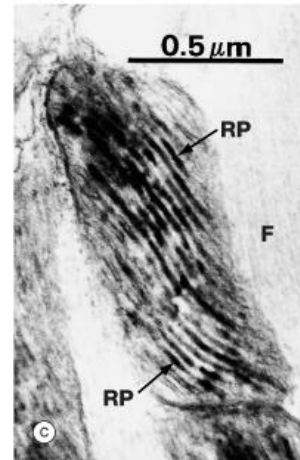
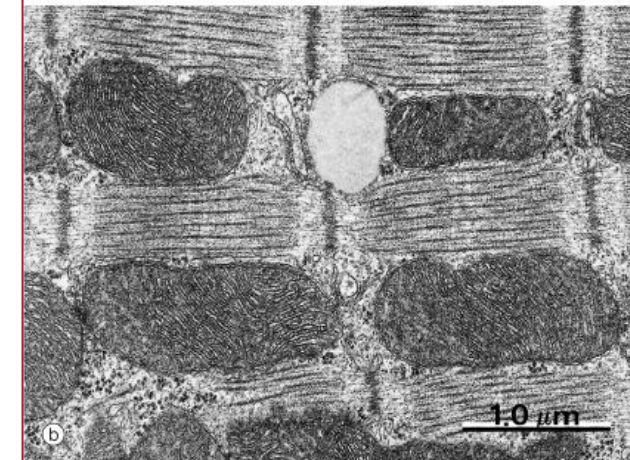
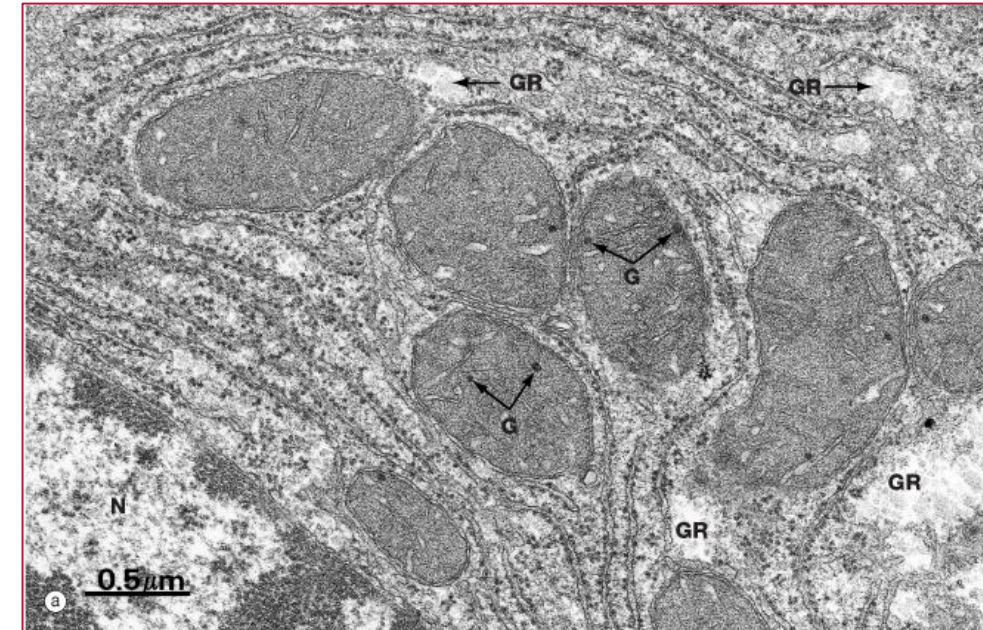
- First LSD characterized
 - Warren Tay (1881) and Bernard Sacs (1896)
- Deficiency of β -hexosaminidase A, α -subunit (HEXA)
- Autosomal recessive disorder
 - Chromosome 15q24.1
- Results in the accumulation of **GM2 ganglioside**
- **Death of neurons in the brain and spinal cord**

Patients present with

Normal birth
Slower growth
Bone and joint deformities
Frequent lung infections and heart disease
Other symptoms

Mitochondria

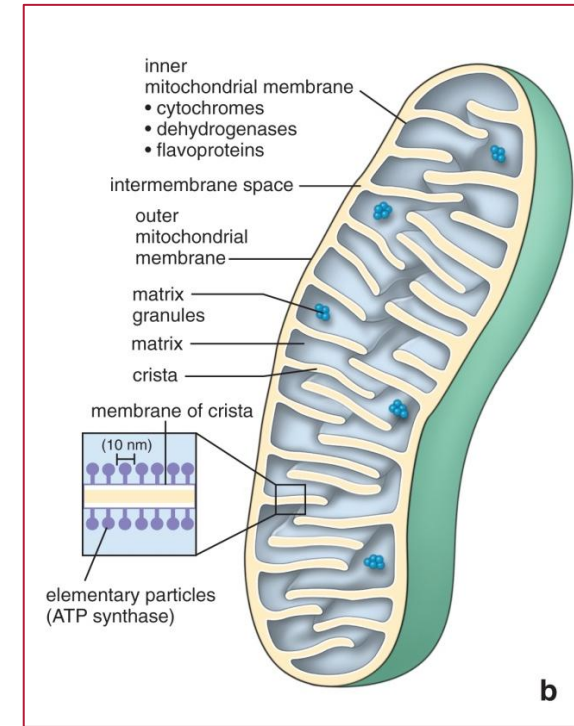
- **Mobile power generators**
 - **Generate ATP**
 - Can divide and multiply independent of the cell cycle
 - Can move about the cell to where they are needed
 - Can change their shape depending on the requirements of the cell
- **Abundant in cells that generate and expend large amounts of energy**
 - Striated muscle cells
- **Evolved from aerobic prokaryotes**
 - Mitochondrial DNA
 - Synthesize own ribosomes



Mitochondria

- **Outer membrane**

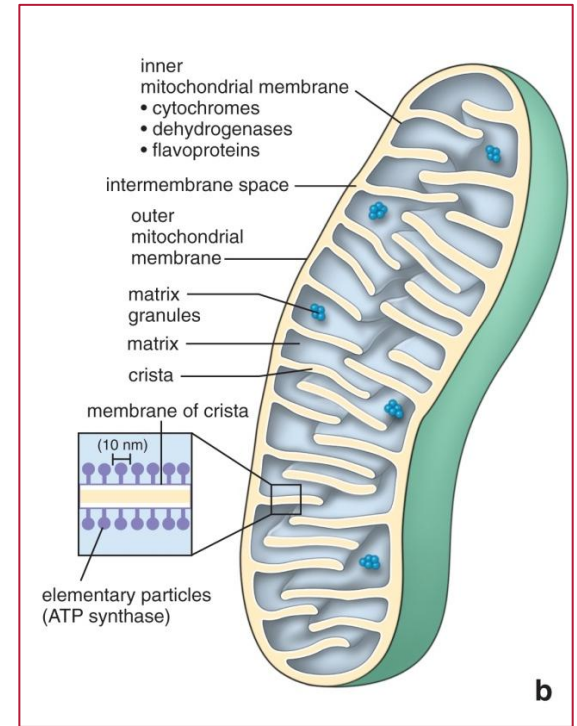
- Contains voltage-dependent anion channels (**mitochondrial porins**)
 - Large channel-forming proteins
 - Permeable to small uncharged molecules (<5kDa)
- Allows for the intermembrane space to be similar to cytoplasm
- Contains several enzymes
 - Phospholipase
 - **Acetyl coenzyme A synthase**
 - **Monoamine oxidase**
- Contains receptors for proteins and polypeptides that translocate into the intermembrane space



Mitochondria

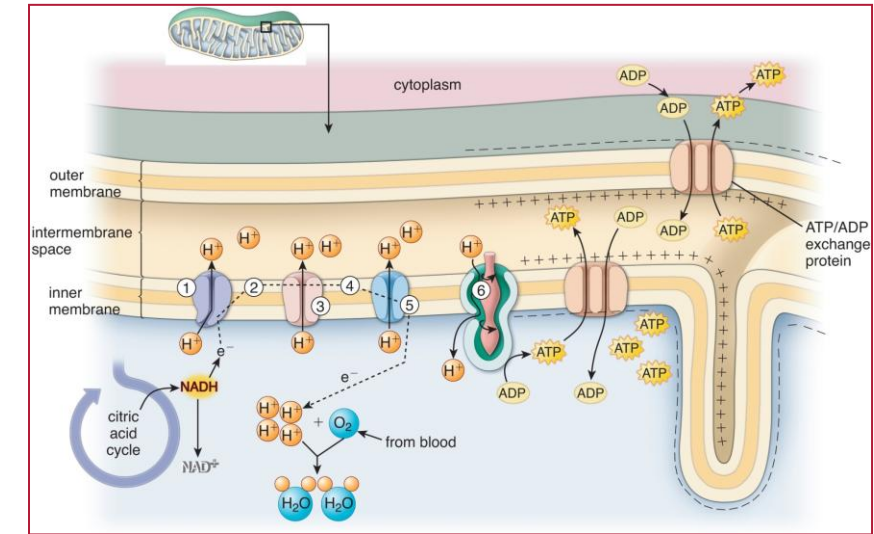
- **Intermembrane space**

- Space between the inner and outer membranes
- Contains specific enzymes that use ATP generated by the inner membrane
 - Creatine kinase
 - Adenylate kinase
 - **Cytochrome c**
 - Important factor in **intrinsic apoptosis pathway**



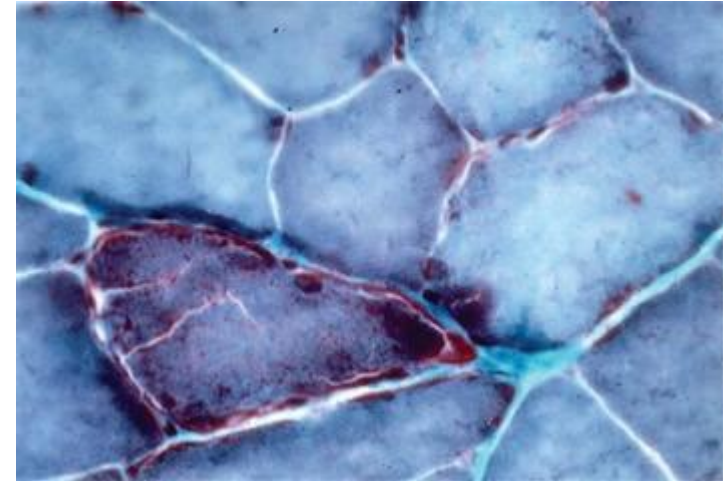
Mitochondria

- **Inner membrane**
 - Thinner and thrown into folds – Cristae
 - **Impermeable to ions**
 - 3 major functions
 - Oxidation reactions: **Respiratory electron transport chain**
 - **Synthesize ATP**
 - Regulate metabolite transport into and out of the matrix
- **Matrix**
 - Contains
 - CO₂ and reduced NADH
 - Dense matrix granules that store Ca²⁺
 - Soluble enzymes of the citric acid cycle (**Krebs cycle**)
 - Enzymes involved in fatty-acid β-oxidation



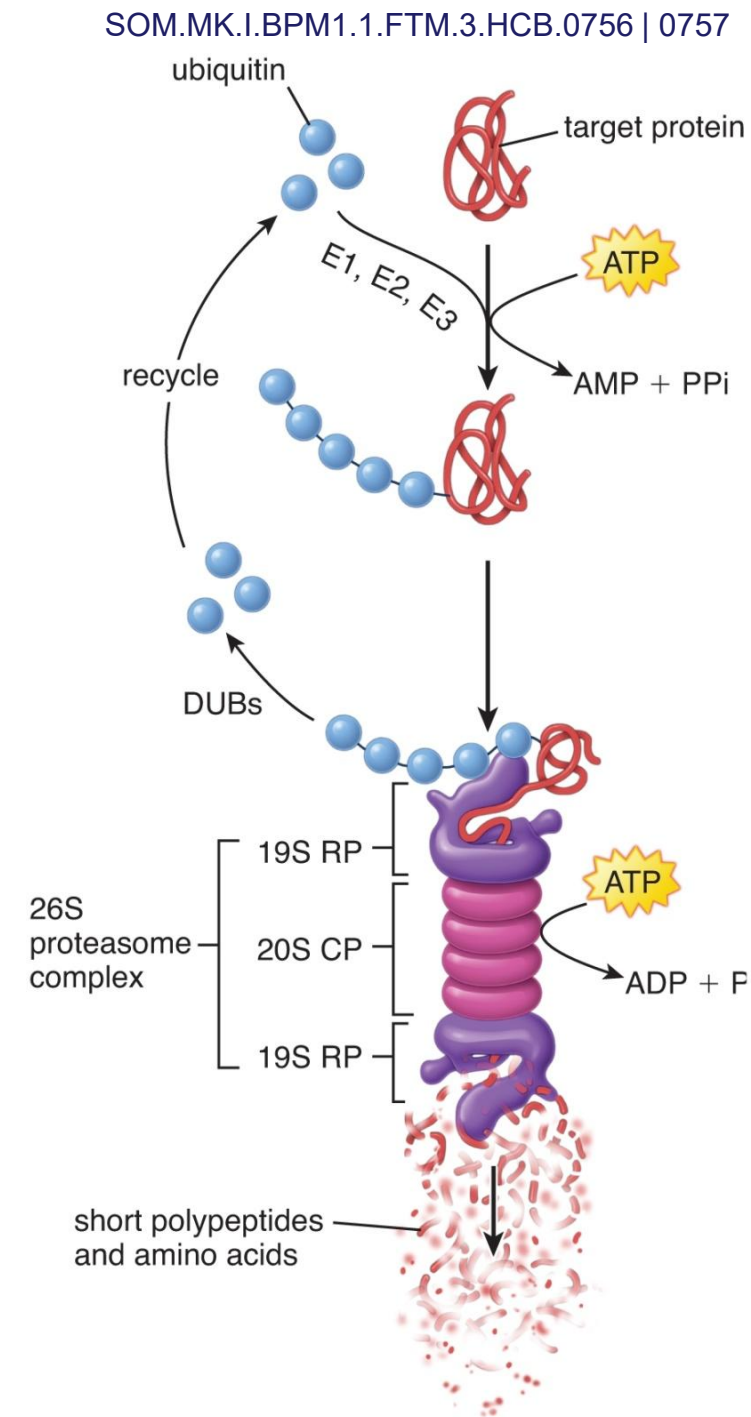
Mitochondrial Diseases

- >40 different disorders
- Mutations in mitochondrial genes
- Results in defects in mitochondrial enzymes required for ATP production
 - Dysfunctional mitochondrial respiratory chain
- Typically affects tissues that require large amounts of ATP
 - e.g. neurons & muscle cells
- **Examples:**
- **Myoclonic epilepsy with ragged red fibers (MERRF)**
 - ~80% of cases due to mutation in the MT-TK tRNA gene
- **Leber hereditary optic neuropathy (LHON)**
 - Mutations in the MT-ND1, MT-ND4, MT-ND4L, or MT-ND6 genes
 - Degeneration of retinal ganglion cells



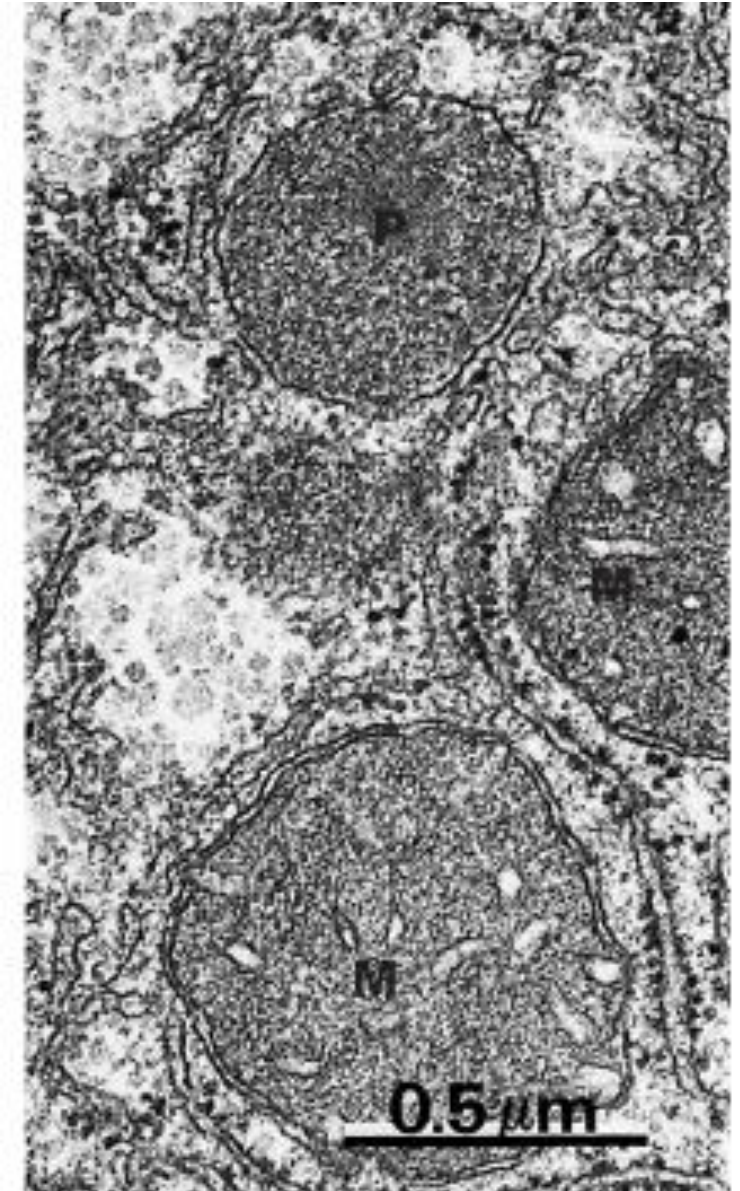
Proteasome

- Proteasome-mediated protein degradation
 - Large protein complex
 - **ATP dependent**
 - **Degrade proteins without involving lysosomes**
- Polyubiquitination
 - Proteins targeted for destruction are covalently tagged with the **ubiquitin** protein
- Destruction of abnormal proteins & normal short-lived regulatory proteins
 - Cyclins, transcription factors, tumor suppressors



Peroxisomes

- Important role in **fat metabolism**
 - **Very long chain fatty acid (VLCFA) beta-oxidation**
 - Catalyze initial reactions in formation of plasmalogens
 - Abundant class of phospholipids in myelin
 - **Deficiencies or abnormal locations can cause profound abnormalities in myelination of nerve cells**
 - e.g. Zellweger's syndrome
- Peroxisomal proteins are synthesized by free (cytoplasmic) ribosomes
 - Proteins must have a peroxisomal targeting signal
- Small organelles (microbodies)
 - 0.5µm diameter
- Specialized to compartmentalize & degrade toxic reactive oxygen molecules
 - Contain **catalase** & other **peroxidases**
 - Converts hydrogen peroxide to O₂ and water
 - Particularly important in hepatocytes
 - Detoxification of ingested alcohol



What comes next? ... Tissues

- Cells with similar embryological origin organize into **tissues with specific biological functions**
- Four basic tissue types
 - **Epithelium:** covers body surfaces, lines body cavities, and forms glands functioning as protection, secretion, absorption and filtration
 - consists of sheets of cells only
 - e.g. epidermis of skin
 - **Connective tissue:** most abundant, underlies or supports the other three basic tissues, both structurally and functionally
 - consists of cells and supporting tissue consisting of proteins and fibers (extracellular matrix – ECM)
 - Has general (e.g. dermis) and specialized (e.g. blood, bone) types based on function
 - **Muscle:** motion and contraction both detectable (e.g. walking) and undetectable (e.g. blood vessel constriction)
 - consists of cells and supporting connective tissue
 - skeletal muscle (biceps in arm), smooth muscle in blood vessels and heart
 - **Nerve:** conducts impulses to control bodily functions
 - Consists of cells and supporting connective tissue

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